We can transmit data in two ways:-

1> Serial - one bit at a time

2> Parallel - >1 bit at a time, on several channels.

Serial Data Transmission:-

- 1 bit at a time over a channel
- high degree of co-operation required b/w communicating devices
- Timing (Rate, duration, timing) must be identical for both transmitter and receiver.

- The receiver must read the incoming signal once every bit interval.

  example:- 1 Mbps Connection
  ↳ bit interval = 1 millionth of a second

- Timing is achieved on both receiver and transmitter using an internal electronic clock.

  ← They must be sychronised

To achieve sychronisation

1> Use a separate clock link
→ enpensive
→ Not viable computer-to-computer
→ Viable ⇝ inside computers (CPU ↔ Memory)

2> Avoid long-term timing issues by sending data in small separate

2.) Avoid any some timing issues by sending data in small separate chunks. this is called Asychronous transmission.

→ send one character at a time

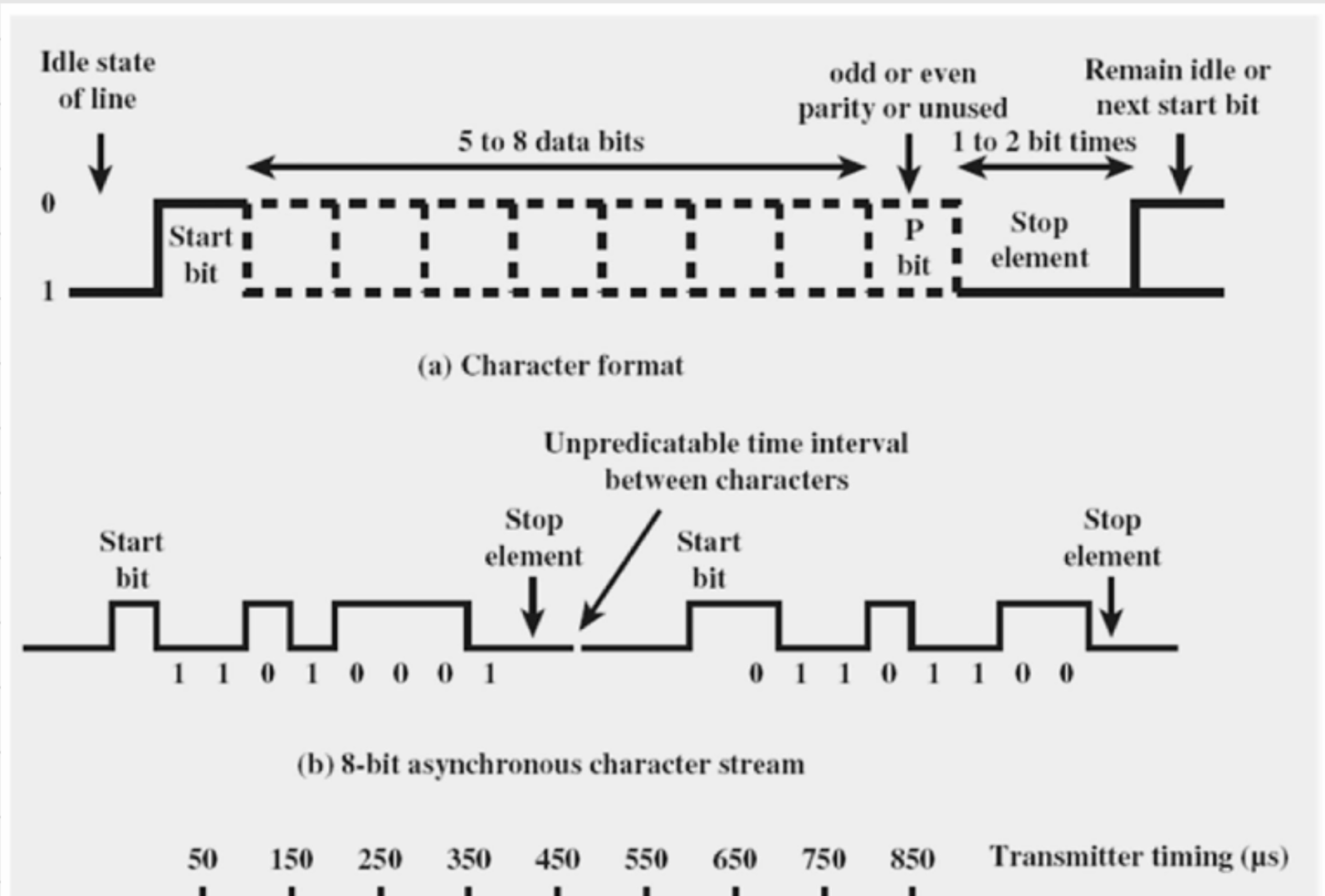
→ each character is wrapped with

- A start bit
- optional parity
- stop bit

3.) Include timing information in the data signal itself so the receiver can stay synchronised. This is called Synchronous transmission.

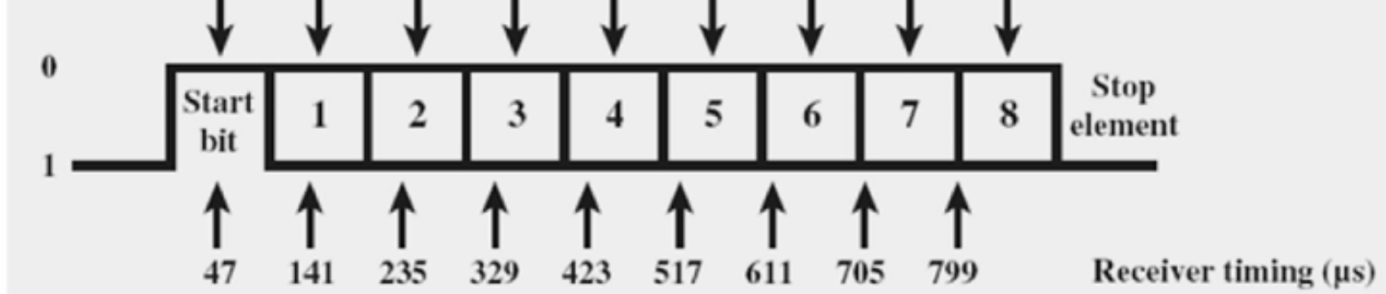
Asynchronous Transmission

→ Timing Mismatch in one bit only affects the current character.



(a) Character format

(b) 8-bit asynchronous character stream



(c) Effect of timing error

Synchronous Transmission

→ Data bits gets transmitted in a $\underbrace{\text{large block of bits}}$

This is called frame.

Opening Flag	Control field	Variable-length data field	Control field	Closing flag

Generic Sychronous Frame Structure

* The receiver identifies the start and end of a frame.
  - ↳ Achieved using a special bit pattern
  - ↳ for preamble (start of frame)
  - ↳ for postamble (end of frame)

* PROBLEM :-

→ More than One Characters are present in the Data Segment.

→ if there is a timing Mismatch of even one bit in any one character, it will Cascade onto the next bits and then to the next character.